

Notes: Krishnamurthy and Muir (JF, 2025)
How Credit Cycles across a Financial Crisis

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1 Big picture

- **Research question.** How do credit markets and output behave around financial crises, and which credit variables predict whether a crisis will be especially severe or likely to occur?
- **Main idea.** A financial crisis is not just a recession with weak output. It is a credit event: credit spreads jump sharply, credit supply tightens, and the real economy contracts for a long time.
- **Core empirical object.** The paper studies credit spreads, credit growth, and real GDP around financial crises across a long international panel. Credit spreads are used as market prices of credit risk and financial-sector losses, while credit growth captures the buildup of financial fragility.
- **Main finding about crisis severity.** The depth of a postcrisis downturn is best described by the interaction between a large rise in spreads and high precrisis credit growth. A big shock is most damaging when the financial sector is fragile.
- **Main finding about crisis prediction.** Crises are often preceded by “frothy” credit-market conditions: credit grows unusually fast while spreads are unusually low. The combination of high credit growth and low spreads predicts financial crises much better than either variable alone.
- **Important distinction.** The change in spreads is more informative for financial crises than the level of spreads. For ordinary recessions, spread levels forecast output, but for financial crises, spread spikes proxy for crisis shocks or credit losses.
- **Theoretical lens.** The evidence supports an FZ view of crises, where the severity of the crisis depends on financial fragility $F_{i,t}$ multiplied by a shock or loss $z_{i,t}$. High fragility amplifies the effect of credit losses.
- **Economic takeaway.** Credit booms are most dangerous when they are accompanied by low credit spreads. Low spreads suggest that credit is cheap and perceived risk is low, but they may also indicate that markets are underpricing crisis risk.

2 Incremental contribution of the paper

- **Adds credit prices to the credit-cycle literature.** Earlier work emphasizes quantities such as credit-to-GDP. This paper adds long-run international data on credit spreads, bringing price information into the analysis of crises.
- **New international spread data.** The authors collect credit-spread data across 17 countries and roughly 150 years, allowing many more crisis episodes than U.S.-only spread data.
- **Separates shock from fragility.** The paper distinguishes the immediate crisis shock, measured by changes in spreads, from precrisis fragility, measured by credit growth.
- **Interaction result.** The key empirical contribution is that spread increases and credit growth interact: spread spikes are especially bad after credit booms.

- **Level versus change.** The paper shows that the change in spreads is the relevant object during financial crises, whereas spread levels are more useful in nonfinancial recessions.
- **Precrisis froth.** The paper documents that spreads are abnormally low before crises even as credit growth is high. This provides evidence consistent with credit-supply expansions and underpricing of risk.
- **Prediction evidence.** A combined froth signal based on low spreads and high credit growth forecasts future financial crises, while it does not strongly predict ordinary recessions.
- **Model discipline.** The findings discipline macrofinance models: models must explain both postcrisis amplification and the precrisis coexistence of rising credit with falling spreads.

3 Data and measurement

3.1 Credit-spread data

The paper constructs a long-run international panel of credit spreads. The early data cover 1869–1929 and come from the *Investors Monthly Manual*, which reports individual bonds traded in London. The authors collect bond prices and bond characteristics, compute country-level yields, and form credit spreads relative to safe benchmarks.

- **Historical sample.** The early sample includes corporate, sovereign, bank, railroad, and other bonds traded in London between 1869 and 1929.
- **Annual spread construction.** Monthly spreads are averaged over the final quarter of each year to obtain an annual spread measure.
- **Composition filter.** The paper requires most bonds in a given country-year to be the same as in the previous year. This reduces spurious spread changes from changes in the bond mix.
- **Modern data.** From 1930 onward, data come from central banks, Global Financial Data, Gilchrist and Mojon for some European countries, and Datastream for recent European crisis countries.
- **Country coverage.** The full spread sample includes 17 countries, including Australia, Canada, France, Germany, Italy, Japan, Sweden, the United Kingdom, and the United States.

Interpretation. Spreads summarize investors’ required compensation for credit risk, default risk, risk premia, and liquidity. Around crises, a large spread increase is interpreted as a sudden repricing of credit losses and risk.

3.2 Crisis dates and turning points

The baseline crisis chronology comes from Jorda, Schularick, and Taylor. For Korea and Greece, the paper uses Laeven and Valencia dates. The authors then modify the dates to align the crisis with the downturn in real GDP.

- **JST crisis dates.** These identify banking crises such as bank runs, closures, or major interventions.

- **Turning points.** The paper applies the Bry-Boschan algorithm to real GDP per capita to identify local peaks and troughs.
- **JST mod. crisis dates.** The baseline analysis dates a crisis as one year after the GDP turning point associated with a crisis.
- **Nonfinancial recessions.** GDP peaks and troughs not associated with banking crises are used as a comparison group.

Interpretation. The modified dating convention is designed to study the economic transition into crisis rather than the exact legal or institutional date of banking distress.

3.3 Output and credit quantities

Real per capita GDP is taken from the Barro-Ursua macroeconomic data. Credit data are from the macrofinancial history database. The main quantity variable is recent credit growth, especially the three-year growth in credit-to-GDP.

- **GDP outcome.** The dependent variable in many regressions is cumulative real per capita GDP growth over one- to five-year horizons.
- **Credit growth.** Credit growth is used as a proxy for financial fragility because credit booms are associated with higher leverage and greater vulnerability to losses.
- **Normalized spreads.** Because spread levels differ across countries and eras, the paper normalizes spreads by country-specific means and also studies alternative normalizations.
- **Spread change.** The main crisis shock variable is the increase in normalized spreads from the prior year to the crisis year.

3.4 Froth and fragility variables

The paper constructs two key indicators.

- **HighCredit.** This variable measures whether recent credit growth has been abnormally high, often based on credit growth being above the median in recent years.
- **HighFroth.** This variable measures whether credit spreads have been abnormally low over the previous five years.
- **Froth interaction.** The product $HighFroth_{i,t} \times HighCredit_{i,t}$ captures episodes in which credit is booming while spreads are low.

Interpretation. High credit growth alone could reflect credit demand or credit supply. High credit growth together with low spreads is more suggestive of a credit-supply expansion.

4 Methodology and empirical specifications

4.1 Event-time analysis around crises

The first empirical exercise plots spreads, credit growth, and GDP around crises. The event date is the modified crisis date. The plots show the average pattern across crises.

- Spreads rise sharply as the economy enters crisis.
- Credit growth is elevated before the crisis and falls afterward.
- GDP declines sharply and remains depressed after the crisis.

Interpretation. Financial crises involve a sharp repricing of credit risk, a credit contraction, and persistent real output losses.

4.2 GDP-growth regressions with crisis interactions

The main output regressions relate future GDP growth to spread variables and credit variables, allowing the relationship to differ between crisis and noncrisis episodes. A representative specification is

$$\Delta \log y_{i,t+k} = a_i + a_t + b'_{crisis} (Z_{i,t} \times \mathbf{1}_{crisis,i,t}) + b'_{non} (Z_{i,t} \times \mathbf{1}_{noncrisis,i,t \rightarrow t+5}) + c' X_{i,t} + \varepsilon_{i,t+k},$$

where $Z_{i,t}$ includes credit spreads, changes in spreads, and credit growth.

- a_i are country fixed effects.
- a_t are year fixed effects.
- $X_{i,t}$ includes lagged GDP-growth controls and other controls used in the paper.
- The coefficients of interest are the interactions with crisis indicators.

Interpretation. The specification asks whether credit variables matter differently in financial crises than in ordinary periods or nonfinancial recessions.

4.3 Fragility times shock

The paper's crisis-severity test is motivated by the idea that crisis losses are amplified by fragility. Empirically, the authors use specifications such as

$$\Delta \log y_{i,t+k} = a_i + a_t + \beta_1 (HighCredit_{i,t} \times \Delta \hat{s}_{i,t}) + \beta_2 HighCredit_{i,t} + \beta_3 \Delta \hat{s}_{i,t} + \dots + \varepsilon_{i,t+k}.$$

A negative β_1 means that increases in spreads are especially bad after high credit growth.

Interpretation. This is the empirical counterpart of the *FZ* mechanism: a large shock is most damaging when it hits a fragile financial system.

4.4 Testing whether spreads are too low before crises

The paper regresses normalized spreads on event-time dummies around crisis dates:

$$\hat{s}_{i,t} = a_i + a_t + \sum_{j=-6}^5 \gamma_j \mathbf{1}_{t=j \text{ relative to crisis}} + u_{i,t}.$$

The authors also construct “fundamental spreads” predicted by macro fundamentals such as lagged GDP growth and credit growth.

Interpretation. If actual spreads are low before a crisis relative to fundamentals, then markets are not simply pricing the fragility implied by the credit boom. This is evidence of froth or underpricing of crisis risk.

4.5 Crisis forecasting

The crisis-prediction specifications ask whether credit-market conditions forecast the occurrence of a crisis over future horizons. A representative version is

$$\Pr(Crisis_{i,t:t+5} = 1) = f(\alpha_i + \beta_1 HighFroth_{i,t} + \beta_2 HighCredit_{i,t} + \beta_3 HighFroth_{i,t} \times HighCredit_{i,t} + \dots).$$

The paper estimates both OLS probability models and logit models.

Interpretation. The central question is whether low spreads and high credit growth jointly predict financial crises, not merely recessions.

4.6 Robustness exercises

The paper reports several robustness checks.

- **Postwar sample.** The authors repeat the main regressions using post-World War II data to check whether the results depend on the early bond data.
- **Alternative crisis dates.** The authors use Reinhart-Rogoff and Baron-Verner-Xiong crisis chronologies.
- **Out-of-sample construction.** The authors construct froth and high-credit measures using information available only up to time t .
- **Alternative spread normalizations.** The paper compares raw spreads, mean-normalized spreads, out-of-sample mean normalizations, z-scores, percentiles, and pre/post split samples.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies robust predictive relationships in a long international panel. It shows that credit-spread changes, credit growth, and their interaction strongly describe crisis severity and that precrisis low spreads combined with high credit growth forecast future financial crises.

The design is not a randomized causal experiment. The evidence is best interpreted as a set of stylized facts and forecasting relationships that are consistent with financial-fragility models.

5.2 Why credit spreads are useful

Credit spreads are forward-looking market prices. They contain information about expected defaults, credit risk premia, and market liquidity. A sudden increase in spreads at the beginning of a crisis is therefore a natural proxy for the shock or loss component $z_{i,t}$.

Key distinction. Spread levels help forecast ordinary output growth because they are forward-looking indicators. But in financial crises, the change in spreads is especially informative because it captures the surprise or loss that hits a fragile financial system.

5.3 Why credit growth proxies for fragility

Credit growth is used as a reduced-form measure of fragility. Rapid credit expansion can raise leverage, increase maturity mismatch, and create vulnerability to losses. The paper does not claim that credit growth is the only form of fragility, but it is observable across countries and over long historical periods.

5.4 Why low spreads plus high credit growth matters

High credit growth can come from high credit demand or high credit supply. Low spreads help distinguish the two. When the quantity of credit rises while the price of credit falls, the pattern is consistent with a credit-supply expansion or a decline in perceived risk.

Economic interpretation. The most dangerous credit booms are not those in which markets are visibly worried. They are booms in which credit expands while spreads look unusually calm.

5.5 Limitations and caveats

- **Crisis dating.** Financial-crisis dates are inherently imperfect. The paper therefore checks alternative chronologies.
- **Historical data quality.** Early spread data require judgment in constructing comparable series across countries and bonds.
- **Measurement of fragility.** Credit growth is an imperfect proxy for financial fragility, but it is available for long historical samples.
- **Forecasting versus causality.** The paper shows strong predictive patterns; the mechanisms are interpreted through theory but not experimentally isolated.
- **Threshold construction.** Variables such as *HighFroth* and *HighCredit* depend on thresholds; the paper checks robustness with out-of-sample threshold construction.

6 Tables and figures

6.1 Tables I and II: Data coverage and crisis dating

Read:

- Table I reports the bond statistics and country-year coverage for the spread data.
- The early bond sample contains 194,854 observations and 4,464 unique bonds.
- Panel B shows the uneven but broad country coverage, with long samples for the United States, United Kingdom, Canada, Australia, Sweden, and Italy.
- Table II lists turning-point dates and crisis dates across countries.
- The bold dates identify the crisis episodes with spread, credit, and output data used in the main analysis.

Economic message: The paper’s contribution depends on combining long-run price data from bond markets with crisis chronologies and macroeconomic outcomes.

Table I
Summary Statistics

This table provides basic summary statistics on the bonds in our sample. The top panel summarizes our historical bond data. The bottom panel documents our coverage across countries and years for the entire sample.

Panel A: Bond Statistics for 1869–1929				
Observations	Unique Bonds	% Gov’t	% Railroad	% Other
194,854	4,464	23%	27%	50%
Median Yield	Median Coupon	Median Discount	Avg Maturity	Median Spread
5.5%	4.2%	6%	17 years	1.9%
Panel B: Full-Sample Coverage by Country				
Country	First Year	Last Year	Total Years	JST Sample
Australia	1869	2011	84	Y
Belgium	1960	2001	42	Y
Canada	1869	2001	113	Y
Denmark	1897	1929	24	Y
France	1869	2009	68	Y
Germany	1927	2021	58	Y
Greece	2003	2012	10	N
Italy	1869	2021	71	Y
Japan	1870	2001	32	Y
Korea	1995	2013	19	N
Netherlands	1880	1929	17	Y
Norway	1876	2003	67	Y
Spain	1871	2021	57	Y
Sweden	1869	2011	79	Y
Switzerland	2001	2022	22	Y
United Kingdom	1869	2014	112	Y
United States	1869	2014	131	Y

Table I: Summary statistics and spread-data coverage.

Table II
Turning Point and Crisis Dates

This table lists the crisis and the corresponding turning point dates from the Bry and Boschan (1971) algorithm for each of the 17 countries for which we have credit spread data. Crisis dates are from JST except for those indicated with *, which are from Laeven and Valencia (2013). In bold we highlight the 40 crisis dates for which we have data on spreads, credit, and output. These dates are the main focus of the study.

Country	Turning Point Dates	Crisis Dates
France	1882, 1929, 2007	1882, 1889, 1930, 2008
Japan	1887, 1898, 1907, 1919, 1925, 1997	1871, 1890, 1901, 1907, 1920, 1927, 1997
Switzerland	1929, 1990, 2008	1870, 1910, 1931, 1991, 2008
United Kingdom	1889, 1973, 1990, 2007	1890, 1974, 1991, 2007
Denmark	1876, 1883, 1920, 1987, 2007	1877, 1885, 1908, 1921, 1987, 2008
Australia	1891, 1894, 1989	1893, 1989
Korea	1996	1997*
Italy	1874, 1887, 1891, 1918, 1929	1873, 1887, 1893, 1907, 1921, 1930, 1935
	1932, 2007	1990, 2008
Netherlands	2008	1921, 2008
Greece	2007	2008*
Belgium	1874, 1883, 1926, 1930, 1937, 2007	1870, 1876, 1885, 1925, 1931, 1934, 1939
		2008
Sweden	1876, 1879, 1904, 1920, 1930, 1990, 2007	1878, 1907, 1922, 1931, 2008
Canada	1907	1907
Norway	1897, 1920, 1923, 1930, 1987	1899, 1922, 1931, 1988
United States	1873, 1892, 1906, 1929, 1981, 2007	1873, 1893, 1907, 1930, 1984, 2007
Germany	1890, 1898, 1928, 2008	1873, 1891, 1901, 1931, 2008
Spain	1884, 1888, 1911, 1925, 1929, 1932, 2007	1883, 1890, 1913, 1924, 1931, 1977, 2008

Table II: Turning-point and crisis dates.

6.2 Figure 1 and Table III: Crisis dynamics and spread normalization

Read Figure 1:

- Spreads rise sharply around the crisis date.
- Credit growth is high before the crisis and falls afterward.
- GDP declines persistently after the crisis date.

Read Table III:

- Table III compares alternative normalizations of credit spreads in one-year GDP-growth regressions.
- Mean-normalized spreads are negatively related to future GDP growth.
- The normalization exercise supports using spreads in a cross-country panel where countries have different average spread levels.

Economic message: Credit spreads are informative about future activity, but the paper must normalize them carefully before pooling countries and eras.

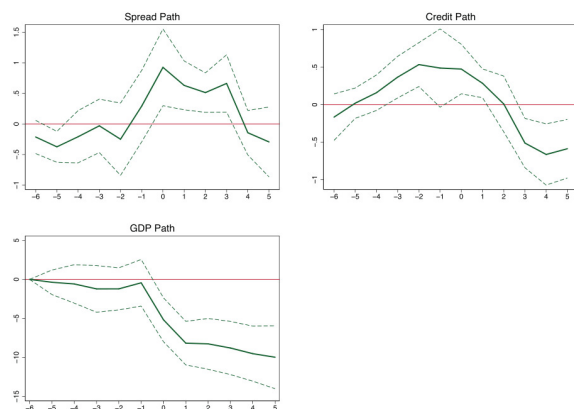


Figure 1. Behavior of spreads, credit, and GDP. This figure plots the behavior of credit spreads, GDP, and the quantity of credit around crises, with date 0 as the JST mod. date associated with the 40 JST crises. GDP is expressed as the percentage fall from year $t - 6$, after demeaning by the growth rate in GDP for country- i . Credit is the three-year growth in credit-to-GDP, demeaned by the country- i growth rate and normalized by the standard deviation of credit growth for the country. Spreads are normalized by dividing by the unconditional mean consistent with our normalization in the main text and plotted as deviations from the country- i mean. See the text for further details. The dashed lines indicate 95% confidence bands. (Color figure can be viewed at wileyonlinelibrary.com)

Figure 1: Spreads, credit, and GDP around crises.

Table III
Credit Spreads and GDP Growth

This table provides regressions of future one-year GDP growth on credit spreads where we consider different normalizations of spreads. The first column uses raw spreads, the second normalizes spreads by dividing by the unconditional mean of the spread in each country, the third also divides by the mean but does so using only information until time $t - 1$ so does not include any look-ahead bias. We refer to this as the out-of-sample (OOS) normalization. The fourth and fifth columns compute a Z-score of spreads and percentile of spreads by country. The last two columns split the sample into high spread (pre-1940) and low spread (post-1940) and examine whether our normalization choice produces similar coefficients across the split. Regressions include a dummy for the early bond data years and both country and year fixed effects. Driscoll-Kraay standard errors with eight lags are in parentheses.

Variables	(1) Raw	(2) MeanNorm	(3) OOSMean	(4) Zscore	(5) Pctile	(6) High Spread	(7) Low Spread
Spread	-0.07 (0.06)						
Lag Spread	0.08 (0.04)						
Spread/Mean		-0.76 (0.19)				-0.67 (0.21)	-0.96 (0.39)
Lag Spread/Mean		0.56 (0.22)				0.54 (0.26)	0.55 (0.36)
Spread/MeanOOS			-0.18 (0.08)				
Lag Spread/OOS			0.04 (0.03)				
Z-score Spread				-0.79 (0.25)			
Lag Z-score Spread				0.56 (0.18)			
Pctile Spread					-1.43 (0.84)		
Lag Pctile Spread					0.47 (0.75)		
Observations	969	968	953	897	897	489	479
R^2	0.37	0.39	0.39	0.36	0.35	0.31	0.68
Number of groups	17	17	17	16	16	13	15

Table III: Credit spreads and GDP growth.

6.3 Figure 2 and Table IV: Spread spikes and crisis severity

Read Figure 2:

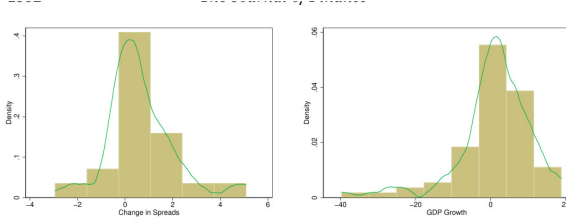
- Crisis episodes display substantial dispersion in both spread changes and subsequent GDP outcomes.
- The scatterplot shows a negative relationship between spread increases and future three-year GDP growth.

Read Table IV:

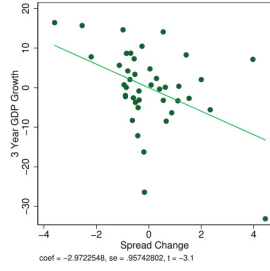
- Table IV estimates GDP-growth regressions with crisis and recession interactions.
- Increases in spreads during financial crises predict worse future GDP growth.
- The crisis interaction is stronger than the corresponding relation for noncrisis dates or ordinary recession dates.

- This table is the main evidence that spread changes are crisis-specific indicators of future output losses.

Economic message: A spread spike is not merely a generic recession signal. It is especially damaging when it marks the transition into a financial crisis.



(A) Distribution of spread changes and future three-year GDP growth around JST mod. crises.



(B) Spread changes against future three-year GDP growth around JST mod. financial crisis dates.

Figure 2. Spreads and GDP around crises. Spread changes and GDP declines around JST mod. crises. (Color figure can be viewed at wileyonlinelibrary.com)

Figure 2: Spread changes and GDP outcomes around crises.

Table IV
Crisis Interactions

This table provides regressions of future cumulative GDP growth $\Delta \ln y_{t+k,i}$, at the three- and five-year horizon, on credit variables. We include interactions of the credit variables with mod. crisis-date or recession-date dummies. Controls include two lags of GDP growth, a dummy for the early bond data years, and both country and year fixed effects. Driscoll-Kraay standard errors with eight lags are in parentheses.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Three-Year GDP Growth									
$\hat{s}_{i,t}$	-1.11 (0.39)								
$\hat{s}_{i,t-1}$	0.81 (0.33)								
$\Delta \hat{s}_{i,t} \times 1_{crisisST,i,t}$	-2.21 (0.74)	-2.11 (0.51)	-2.15 (0.56)						
$\Delta cred_{i,t} \times 1_{crisisST,i,t}$		-2.06 (0.95)	-2.10 (0.84)						
$1_{crisisST,i,t}$			0.19 (1.27)						
$\hat{s}_{i,t} \times 1_{crisisST,i,t}$				-0.95 (0.51)					
$\hat{s}_{i,t} \times 1_{nocrisis,i,t \rightarrow t+5}$					-0.52 (0.52)				
$\Delta \hat{s}_{i,t} \times 1_{nocrisis,i,t \rightarrow t+5}$						-1.03 (0.54)			
$\hat{s}_{i,t} \times 1_{recess,i,t}$							-2.22 (0.65)	-1.70 (0.72)	
$\Delta \hat{s}_{i,t} \times 1_{recess,i,t}$								-1.17 (0.88)	
Observations	826	826	826	826	839	839	826	839	826
R^2	0.53	0.53	0.54	0.54	0.52	0.52	0.53	0.54	0.55
Number of groups	15	15	15	15	15	15	15	15	15
Panel B: Five-Year GDP Growth									
$\hat{s}_{i,t}$	-1.04 (0.43)								
$\hat{s}_{i,t-1}$	1.38 (0.31)								
$\Delta \hat{s}_{i,t} \times 1_{crisisST,i,t}$		-1.48 (0.60)	-1.35 (0.28)	-1.32 (0.42)					
$\Delta cred_{i,t} \times 1_{crisisST,i,t}$			-2.65 (1.10)	-2.62 (0.92)					
$1_{crisisST,i,t}$				-0.13 (1.46)					
$\hat{s}_{i,t} \times 1_{crisisST,i,t}$					-0.38 (0.36)				
$\hat{s}_{i,t} \times 1_{nocrisis,i,t \rightarrow t+5}$						-0.14 (0.56)			

(Continued)

Table IV, part 1: Crisis interactions.

6.4 Table IV continued and Table V: Crisis effects across the distribution

Read Table IV continued:

- The continuation completes the five-year GDP-growth panel.
- The results show that crisis spread increases are associated with prolonged output losses.

Read Table V:

- Table V uses quantile regressions of future output growth.
- Spread increases are especially informative for lower quantiles of GDP growth.
- This means spread spikes mainly predict downside macroeconomic tail risk.

Economic message: Credit spreads help identify the left-tail macro risk that makes financial crises unusually severe.

Table IV—Continued

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel B: Five-Year GDP Growth									
$\Delta \hat{s}_{i,t} \times 1_{\text{HOCrisis},i,t \rightarrow t+5}$								−0.86 (0.38)	
$\hat{s}_{i,t} \times 1_{\text{Recess},i,t}$								−1.69 (0.72)	−0.93 (1.02)
$\Delta \hat{s}_{i,t} \times 1_{\text{Recess},i,t}$								−1.74 (1.42)	816
Observations	816	816	816	816	829	829	816	829	816
R^2	0.52	0.52	0.52	0.52	0.51	0.51	0.52	0.52	0.53
Number of groups	15	15	15	15	15	15	15	15	15

Table IV, part 2: Continued five-year GDP growth results.

Table V
Quantile Regressions

We run quantile regressions of future output growth over the next year on the change in credit spreads and the three-year growth in credit/GDP for different quantiles. Controls include two lags of GDP growth, a dummy for the early bond data years, and country and time fixed effects. Our main result is that increases in spreads are particularly informative for lower quantiles of GDP growth. Standard errors in parentheses clustered by year.

Variables	(1) 90 th	(2) 75 th	(3) 50 th	(4) 25 th	(5) 10 th
$\Delta s_{i,t}$	−0.39 (0.29)	−0.36 (0.46)	−0.59 (0.19)	−1.12 (0.57)	−0.55 (0.83)
$\Delta \text{Credit}_{i,t}$	−1.25 (0.25)	−1.05 (0.23)	−1.00 (0.37)	−0.85 (0.45)	−0.77 (0.56)
Observations	826	826	826	826	826
R^2	0.30	0.41	0.50	0.47	0.42

Table V: Quantile regressions.

6.5 Tables VI and VII: Fragility interaction and low precrisis spreads

Read Table VI:

- Table VI asks which spread crises turn out badly.
- Panel A interacts spread changes with the *HighCredit* variable.
- Panel B uses the interaction between credit growth and spread changes directly.
- The results show that spread increases are most damaging after high credit growth.

Read Table VII:

- Table VII examines whether spreads are unusually low before crises.
- Spreads tend to be low in the years leading up to crises and high near the crisis date.
- The result remains when controlling for macro fundamentals.

Economic message: Crises are preceded by a dangerous combination: credit-market calm and credit expansion, followed by a sharp repricing.

Table VI
Which Spread Crises Turn Out Badly?

We run regressions where the left-hand side is GDP growth at various horizons. In Panel A, the right-hand side contains a variable *HighCredit* which counts the number of times that credit growth has been above median in each of the past three years. Panel B instead directly interacts changes in credit spreads with credit growth over the previous three years. The table shows that an increase in spreads is negatively associated with subsequent output growth if lagged credit growth has also been high. Controls include two lags of GDP growth, a dummy for the early bond data years, and both time and country fixed effects. Driscoll-Kraay standard errors with eight lags are in parentheses.

Panel A					
Variables	(1) 1yr	(2) 2yr	(3) 3yr	(4) 4yr	(5) 5yr
$(HighCredit_{i,t}) \times \Delta s_{i,t}$	-0.94 (0.27)	-1.45 (0.45)	-1.96 (0.62)	-1.72 (0.63)	-1.39 (0.42)
$(HighCredit_{i,t}) \times s_{i,t-1}$	-0.32 (0.15)	-0.64 (0.34)	-0.68 (0.52)	-0.69 (0.56)	-0.32 (0.56)
$(1 - HighCredit_{i,t}) \times s_{i,t}$	-0.52 (0.34)	-0.50 (0.27)	0.05 (0.30)	0.01 (0.32)	-0.25 (0.58)
$(1 - HighCredit_{i,t}) \times s_{i,t-1}$	0.35 (0.30)	0.56 (0.38)	0.36 (0.33)	0.80 (0.41)	1.47 (0.35)
<i>HighCredit</i> _{<i>i,t</i>}	-0.31 (0.42)	-0.96 (0.83)	-1.25 (1.04)	-0.94 (1.32)	-0.27 (1.86)
Observations	835	830	825	820	815
<i>R</i> ²	0.45	0.54	0.55	0.54	0.53
Number of groups	15	15	15	15	15

Panel B					
Variables	(1) 1yr	(2) 2yr	(3) 3yr	(4) 4yr	(5) 5yr
$\Delta Credit_{i,t} \times \Delta s_{i,t}$	-0.20 (0.15)	-0.29 (0.20)	-0.38 (0.23)	-0.23 (0.22)	0.02 (0.13)
$\Delta s_{i,t}$	-0.69 (0.24)	-0.97 (0.32)	-1.00 (0.35)	-0.97 (0.34)	-1.01 (0.26)
$\Delta Credit_{i,t}$	-0.19 (0.14)	-0.66 (0.26)	-1.12 (0.35)	-1.37 (0.42)	-1.39 (0.54)
Observations	836	831	826	821	816
<i>R</i> ²	0.45	0.54	0.55	0.54	0.53
Number of groups	15	15	15	15	15

Table VI: Which spread crises turn out badly?

Table VII
Are Spreads before a Crisis Too Low?

We run regressions of our normalized spreads on year dummies from $t - 6$ to $t + 5$ where t is the date of a mod. crisis. Columns (1) and (3) include country fixed effects but not time fixed effects, while the rest include both. We also consider dummies interacted with severe and mild crises, recessions, and credit growth. Controls include the dummy for the early bond data years and the five-year lagged spreadnorm, which controls for variation in the level of spreads. Driscoll-Kraay standard errors with eight lags are in parentheses.

Precrisis Spreads								
Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
1_{t+5}	-0.29 (0.28)	-0.12 (0.23)	-0.31 (0.29)	-0.13 (0.23)	-0.10 (0.22)	-0.11 (0.24)	-0.13 (0.23)	0.03 (0.18)
1_{t+4}	-0.14 (0.18)	-0.11 (0.22)	-0.15 (0.18)	-0.11 (0.21)	-0.05 (0.22)	-0.08 (0.24)	-0.08 (0.25)	0.02 (0.20)
1_{t+3}	0.67 (0.24)	0.34 (0.29)	0.65 (0.24)	0.33 (0.28)	0.38 (0.30)	0.35 (0.29)	0.49 (0.35)	-0.00 (0.16)
1_{t+2}	0.52 (0.16)	0.26 (0.19)	0.50 (0.16)	0.25 (0.19)	0.28 (0.19)	0.27 (0.19)	0.33 (0.22)	0.04 (0.11)
1_{t+1}	0.63 (0.20)	0.39 (0.24)	0.63 (0.20)	0.38 (0.24)	0.40 (0.25)	0.40 (0.24)	0.44 (0.26)	0.18 (0.11)
1_t	0.93 (0.32)	0.53 (0.40)	0.93 (0.32)	0.50 (0.40)	0.52 (0.41)	0.56 (0.41)	0.68 (0.44)	0.19 (0.11)
1_{t-1}	0.30 (0.30)	0.09 (0.41)	0.29 (0.30)	0.08 (0.42)	0.06 (0.44)	0.09 (0.42)	0.17 (0.46)	0.16 (0.12)
1_{t-2}	-0.25 (0.30)	-0.51 (0.59)	-0.25 (0.30)	-0.50 (0.58)	-0.52 (0.60)	-0.50 (0.58)	-0.40 (0.65)	0.03 (0.09)
1_{t-3}	-0.04 (0.21)	-0.29 (0.27)	-0.04 (0.21)	-0.28 (0.27)	-0.25 (0.27)	-0.26 (0.26)	-0.18 (0.27)	0.00 (0.14)
1_{t-4}	-0.21 (0.21)	-0.41 (0.22)						
1_{t-5}	-0.37 (0.13)	-0.55 (0.18)						
1_{t-6}	-0.21 (0.13)	-0.37 (0.15)						
$1_{t-6,t-4}$			-0.26 (0.14)	-0.44 (0.15)				
$1_{t-6,t-4} \times severe$					-0.42 (0.17)			
$1_{t-6,t-4} \times mild$						-0.38 (0.20)		
$1_{t-6,t-4} \times \Delta credit_t$							-0.34 (0.27)	
<i>recession</i> _{$t-6,t-4$}								-0.03 (0.13)
Observations	799	799	802	802	791	791	715	791
<i>R</i> ²	0.13	0.48	0.13	0.48	0.47	0.47	0.49	0.45
Number of groups	15	15	16	16	14	14	14	14
Year FE	N	Y	N	Y	Y	Y	Y	Y

Table VII: Are spreads before a crisis too low?

6.6 Figure 3 and Table VIII: Froth and crisis prediction

Read Figure 3:

- Actual spreads are below fundamental spreads before crises.
- Actual spreads jump above fundamentals during the crisis and then fall back.
- The visual pattern supports the idea that markets underprice crisis risk during the boom.

Read Table VIII:

- Table VIII forecasts crises using *HighFroth*, *HighCredit*, and their interaction.
- The interaction between froth and high credit growth is strongly positive for crisis prediction.
- The same variables do not predict ordinary recessions as strongly.

Economic message: Credit froth is not just low spreads or high credit growth separately. It is the combination of both.

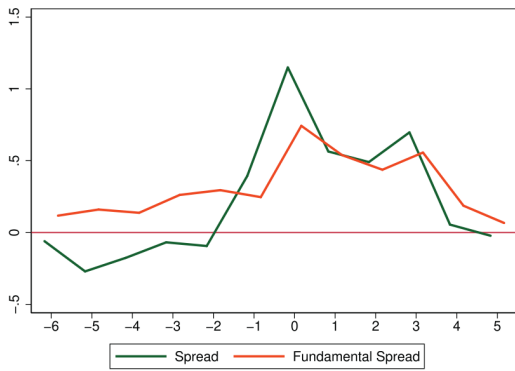


Figure 3. Spreads and fundamental spreads. This figure plots the path of spreads and fundamental spreads in the years surrounding a JST mod. crisis. The paths are formed by running regressions with dummies at various dates. “Fundamental spreads” are computed as the predicted value from a regression of spreads on fundamentals including two lags of GDP growth and the change in credit. (Color figure can be viewed at wileyonlinelibrary.com)

fitted spread represents a “fundamental” spread based on the relation between

Figure 3: Actual spreads and fundamental spreads.

Table VIII
Credit Market Froth and Fragility

We explore whether low spreads and high credit growth can forecast crises. $HighFroth_{i,t}$ measures whether spreads have been abnormally low in the last five years. $HighCredit_{i,t}$ measures whether credit growth has been abnormally high in the last three years. See text for details. Panel A uses these variables to forecast a financial crisis (using JST mod. dates). We run regressions on the cumulative crisis indicator at the five-year horizon (e.g., we predict whether a crisis occurs in any of the next five years). We also interact $HighFroth_{i,t}$ with $HighCredit_{i,t}$, as this captures episodes where credit is booming and spreads are falling. Panel B repeats this exercise for recessions. We include country fixed effects and a dummy for the early bond data years. Driscoll-Kraay standard errors with eight lags are in parentheses.

Panel A: Predicting Crises OLS					
Variables	(1)	(2)	(3)	(4)	(5) Rest. Sample
$(HighFroth)_{i,t}$	0.21 (0.06)			0.04 (0.11)	
$(HighCredit)_{i,t}$		0.18 (0.05)		-0.00 (0.09)	
$(HighFroth) \times (HighCredit)_{i,t}$			0.27 (0.09)	0.24 (0.20)	0.29 (0.09)
Observations	657	848	598	598	545
R^2	0.09	0.10	0.12	0.12	0.12
Number of groups	14	15	14	14	12

Panel B: Predicting Recessions OLS					
Variables	(1)	(2)	(3)	(4)	(5) Rest. Sample
$(HighFroth)_{i,t}$	0.02 (0.08)			0.18 (0.12)	
$(HighCredit)_{i,t}$		-0.05 (0.08)		0.13 (0.14)	
$(HighFroth) \times (HighCredit)_{i,t}$			0.09 (0.08)	-0.16 (0.23)	0.06 (0.08)
Observations	657	848	598	598	545
R^2	0.21	0.15	0.22	0.23	0.24
Number of groups	14	15	14	14	12

Table VIII: Credit-market froth and fragility.

6.7 Table IX and Figure 4: Logit evidence and conditional probabilities

Read Table IX:

- Table IX repeats the crisis-prediction analysis in a logit model.
- The interaction of $HighFroth$ and $HighCredit$ continues to predict crises.
- The logit results support the probability interpretation of the froth signal.

Read Figure 4:

- The figure plots conditional probabilities of a crisis at one- to five-year horizons.
- The probability is highest when both high froth and high credit are present.
- Conditioning on only high credit or only high froth produces smaller probabilities.

Economic message: The most dangerous state is not merely a credit boom; it is a credit boom priced as if risk is low.

Table IX
Credit Market Froth and Fragility: Logit

We explore whether low spreads and high credit growth can forecast GDP downturns and crises. $HighFroth_{i,t}$ measures whether spreads have been abnormally low in the last five years. $HighCredit_{i,t}$ measures whether credit growth has been abnormally high in the last three years. See text for further details. Panel A uses these variables to forecast a financial crisis (using JST mod. dates). We run Logit regressions on the cumulative crisis indicator at the five-year horizon (e.g., we predict whether a crisis occurs in any of the next five years). We also interact $HighFroth_{i,t}$ and $HighCredit_{i,t}$, as this captures episodes where credit is booming and spreads are falling. Panel B repeats this exercise for recessions. We include country fixed effects and a dummy for the early bond data years. Standard errors are double-clustered by country and year.

Panel A: Predicting Crises Logit				
Variables	(1)	(2)	(3)	(4)
$(HighFroth)_{i,t}$	1.55 (0.93)			0.55 (1.57)
$(HighCredit)_{i,t}$		1.28 (0.46)		0.29 (1.52)
$(HighFroth) \times (HighCredit)_{i,t}$			1.99 (0.86)	1.34 (2.06)
Observations	604	798	545	545
Pseudo R^2	0.0870	0.0963	0.109	0.111

Panel B: Predicting Recessions Logit				
Variables	(1)	(2)	(3)	(4)
$(HighFroth)_{i,t}$	0.09 (0.45)			1.00 (0.72)
$(HighCredit)_{i,t}$		-0.20 (0.47)		0.75 (0.80)
$(HighFroth) \times (HighCredit)_{i,t}$			0.54 (0.46)	-0.88 (0.92)
Observations	657	848	595	595
Pseudo R^2	0.170	0.120	0.176	0.181

Table IX: Credit-market froth and fragility in logit regressions.

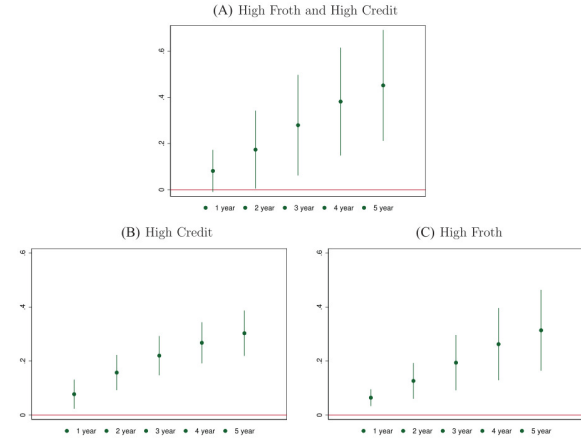


Figure 4. Conditional crisis probabilities. We plot the cumulative probability of a crisis at each horizon when we condition on credit market conditions. $HighFroth_{i,t}$ is a dummy equal to 1 if abnormal credit spreads (the residual of spreads regressed on two lags of GDP and credit growth) over the last five years have been below their median. $HighCredit_{i,t}$ is a dummy equal to 1 if credit growth over the past three years has been high. We compare the future probability of a crisis when credit markets appear “frothy” ($HighFroth_{i,t}$ and $HighCredit_{i,t}$ both equal 1) to when they are not (both equal to zero). In the bottom panel, we compare this to only cases where credit is high. See text for more details. Results are from a Logit model with country fixed effects with standard errors double-clustered by year and country. (Color figure can be viewed at wileyonlinelibrary.com)

Figure 4: Conditional crisis probabilities.

6.8 Tables X and XI: Out-of-sample and postwar checks

Read Table X:

- Table X constructs froth and credit indicators using information available only up to time t .
- The out-of-sample signals are weaker than the full-sample signals.
- The interaction still points in the same broad direction for crisis prediction.

Read Table XI:

- Table XI repeats the spreads-and-GDP analysis using postwar data.
- The core spread-crisis relationship remains visible in the postwar sample.
- Standard errors rise because the postwar sample has fewer historical crisis episodes.

Economic message: The central patterns are not purely artifacts of the full historical sample, although predictive power is naturally weaker in stricter out-of-sample exercises.

Table X
Out-of-Sample Results: Credit Market Froth and Fragility

Our previous regressions use the full sample to determine cutoffs for high froth in credit spreads and high credit growth. We repeat our results where we use out-of-sample measures of froth and high credit growth episodes using only the information up to time t to construct each variable. See Table VIII for details. Standard errors are double-clustered by country and year.

Panel A: Predicting Crises OLS				
Variables	(1)	(2)	(3)	(4)
$(HighFroth)_{i,t}$	0.05 (0.08)			0.00 (0.09)
$(HighCredit)_{i,t}$		0.14 (0.06)		0.09 (0.10)
$(HighFroth)_{i,t} \times (HighCredit)_{i,t}$			0.14 (0.07)	0.06 (0.13)
Observations	624	699	620	620
R^2	0.08	0.10	0.10	0.10
Number of groups	15	15	15	15

Panel B: Predicting Recessions OLS				
Variables	(1)	(2)	(3)	(4)
$(HighFroth)_{i,t}$	-0.05 (0.10)			0.02 (0.18)
$(HighCredit)_{i,t}$		-0.07 (0.08)		0.04 (0.12)
$(HighFroth)_{i,t} \times (HighCredit)_{i,t}$			-0.08 (0.08)	-0.12 (0.20)
Observations	624	699	620	620
R^2	0.13	0.12	0.12	0.13
Number of groups	15	15	15	15

Table X: Out-of-sample froth and fragility.

Table XI
Postwar Data: Spreads and GDP

This table provides regressions of future cumulative GDP growth $\Delta \ln y_{t+k,j}$ on credit spreads at the three-year horizon. We include interactions with crisis or recession dummies. Controls include two lags of GDP growth, a dummy for the early bond data years, and both country and year fixed effects. Driscoll-Kraay standard errors with eight lags are in parentheses.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	1	2	3	4	5	6	7	8	9
$\hat{s}_{i,t}$	-0.80 (0.32)								
$\hat{s}_{i,t-1}$	0.63 (0.51)								
$\Delta \hat{s}_{i,t} \times 1_{crisisST,i,t}$		-1.43 (0.53)	-1.25 (0.56)	-2.02 (0.37)					
$\Delta cred_{i,t} \times 1_{crisisST,i,t}$			-0.58 (0.55)	-1.39 (0.35)					
$1_{crisisST,i,t}$				3.02 (0.70)					
$\hat{s}_{i,t} \times 1_{crisisST,i,t}$					-1.06 (0.57)				
$\hat{s}_{i,t} \times 1_{nocrisis,i,t \rightarrow t+5}$						-0.03 (0.50)			
$\Delta \hat{s}_{i,t} \times 1_{nocrisis,i,t \rightarrow t+5}$							-0.54 (0.40)		
$\hat{s}_{i,t} \times 1_{recess,i,t}$								-0.82 (0.73)	-0.76 (0.72)
$\Delta \hat{s}_{i,t} \times 1_{recess,i,t}$								0.51 (0.64)	
Observations	447	447	447	447	460	460	447	460	447
R^2	0.71	0.71	0.71	0.71	0.68	0.68	0.70	0.68	0.71
Number of groups	13	13	13	13	13	13	13	13	13

Table XI: Postwar spreads and GDP.

6.9 Tables XII and XIII: Postwar severity and prediction

Read Table XII:

- Table XII asks, in postwar data, when spread crises turn out badly.
- The interaction of high credit and spread changes remains negatively related to future GDP growth.
- The evidence is consistent with the full-sample fragility-times-shock interpretation.

Read Table XIII:

- Table XIII repeats the froth-and-fragility crisis-prediction regressions in postwar data.
- High credit and low spreads continue to contain crisis-prediction information.
- The evidence is less precise than in the full sample but directionally consistent.

Economic message: The paper's main mechanisms appear in both long-run and postwar samples.

Table XII
Postwar Data: Which Spread Crises Turn Out Badly?

We run regressions where the left-hand side is GDP growth at various horizons. The right-hand side contains a variable *HighCredit* which counts the number of times that credit growth has been above median in each of the past three years. We include interactions of this variable with spread changes and the level of spreads. The table shows that an increase in spreads is negatively associated with subsequent output growth if credit growth has also been high. Controls include two lags of GDP growth, country fixed effects, time fixed effects, and a dummy for the early bond data years. Driscoll-Kraay standard errors with eight lags are in parentheses.

When is an Increase in Spreads Particularly Bad for GDP?					
Variables	(1) 1yr	(2) 2yr	(3) 3yr	(4) 4yr	(5) 5yr
$(HighCredit_{i,t}) \times \Delta s_{i,t}$	-0.10 (0.24)	-0.35 (0.29)	-0.83 (0.29)	-0.53 (0.37)	-0.44 (0.62)
$(HighCredit_{i,t}) \times s_{i,t-1}$	-0.38 (0.22)	-0.92 (0.36)	-1.06 (0.47)	-0.96 (0.64)	-0.86 (0.87)
$(1 - HighCredit_{i,t}) \times s_{i,t}$	-0.98 (0.30)	-0.76 (0.52)	-0.51 (0.65)	0.06 (0.68)	0.24 (0.87)
$(1 - HighCredit_{i,t}) \times s_{i,t-1}$	0.91 (0.23)	1.09 (0.46)	1.41 (0.69)	1.40 (0.89)	1.41 (0.90)
<i>HighCredit</i> _{<i>i,t</i>}	0.19 (0.48)	0.36 (0.75)	0.27 (1.00)	-0.09 (1.17)	-0.50 (1.40)
Observations	457	452	447	442	437
<i>R</i> ²	0.72	0.71	0.72	0.75	0.76
Number of groups	13	13	13	13	13

Table XII: Postwar spread crises and GDP losses.

Table XIII
Postwar Data: Credit Market Froth and Fragility

We explore whether low spreads and high credit growth can forecast crises. *HighFroth*_{*i,t*} measures whether spreads have been abnormally low in the last five years. *HighCredit*_{*i,t*} measures whether credit growth has been abnormally high in the last three years. See text for details. Panel A uses these variables to forecast a financial crisis (using JST mod. dates). We run regressions on the cumulative crisis indicator at the five-year horizon (e.g., we predict whether a crisis occurs in any of the next five years). We also interact *HighFroth*_{*i,t*} with *HighCredit*_{*i,t*}, as this captures episodes where credit is booming and spreads are falling. Panel B repeats this exercise for recessions. We include country fixed effects and a dummy for the early bond data years. Driscoll-Kraay standard errors with eight lags are in parentheses.

Panel A: Predicting Crises OLS				
Variables	(1)	(2)	(3)	(4)
$(HighFroth)_{i,t}$	0.06 (0.07)			-0.04 (0.13)
$(HighCredit)_{i,t}$		0.18 (0.05)		0.13 (0.09)
$(HighFroth) \times (HighCredit)_{i,t}$			0.19 (0.07)	0.09 (0.16)
Observations	310	470	310	310
<i>R</i> ²	0.12	0.12	0.14	0.15
Number of groups	10	13	10	10

Panel B: Predicting Recessions OLS				
Variables	(1)	(2)	(3)	(4)
$(HighFroth)_{i,t}$	0.16 (0.07)			0.21 (0.13)
$(HighCredit)_{i,t}$		-0.00 (0.11)		0.16 (0.18)
$(HighFroth) \times (HighCredit)_{i,t}$			0.18 (0.12)	-0.14 (0.26)
Observations	310	470	310	310
<i>R</i> ²	0.10	0.13	0.11	0.11
Number of groups	10	13	10	10

Table XIII: Postwar credit-market froth and fragility.

6.10 Tables XIV and XV: Forecast stability and alternative crisis dates

Read Table XIV:

- Table XIV compares postwar forecast errors using coefficients estimated from prewar data.
- Models using spreads and the spread-credit interaction perform well relative to baselines.
- The table checks the stability of the empirical relationships across eras.

Read Table XV:

- Table XV repeats the crisis-spread interaction using alternative crisis dates from Baron-Verner-Xiong and Reinhart-Rogoff.
- The spread-spike coefficient remains negative and economically similar across chronologies.
- This reduces concern that the main Table IV result is only a feature of the JST dating convention.

Economic message: The spread-spike result is robust to different ways of defining financial crises.

Table XIV
Predicting Output Out-of-Sample

We use prewar data to fit our regressions, then compare the mean absolute forecast errors out-of-sample using postwar data keeping the coefficients fixed from the earlier exercise. Numbers given as a percent. There are 447 postwar observations across the countries in our sample. "Baseline" uses two lags of GDP growth, country fixed effects, and a dummy for the early bond data years. "Spread" uses only data from spreads in the regression, "Credit" uses only credit growth. "Both" uses spreads and credit individually, and "Interaction" uses the interaction term between credit and spreads. In Panel A, credit is a dummy for being above its mean, while in Panel B we take the growth in credit as a continuous measure.

Panel A: High Credit Dummy				
Baseline	Spread	Credit	Both	Interaction
6.16	5.81	6.32	5.87	5.22
Panel B: Continuous Change in Credit				
Baseline	Spread	Credit	Both	Interaction
6.16	5.87	6.58	6.29	6.10

Table XIV: Predicting output out-of-sample.

Table XV
Crisis Interactions, Alternative Dating

Crisis and Spread Interaction using Baron, Verner, and Xiong (2021) (BVX) and Reinhart and Rogoff (2009) (RR) dates and the Bry and Boschan (1971) algorithm to date the start of the crisis recession. The left-hand-side variable is future three-year GDP growth. Controls are the same as in Table IV. Driscoll-Kraay standard errors with eight lags are in parentheses.

Variables	(1) JST	(2) BVX	(3) RR
$\Delta \hat{s}_{i,t} \times 1_{\text{crisisST},i,t}$	-2.21 (0.74)		
$\Delta \hat{s}_{i,t} \times 1_{\text{crisisBVX},i,t}$		-2.47 (0.71)	
$\Delta \hat{s}_{i,t} \times 1_{\text{crisisRR},i,t}$			-2.36 (0.48)
Observations	826	826	826
R^2	0.53	0.54	0.53
Number of groups	15	15	15

Table XV: Crisis interactions with alternative dating.

6.11 Tables XVI and XVII: Alternative dates for froth and prediction

Read Table XVI:

- Table XVI revisits the low-precrisis-spread evidence using alternative crisis chronologies.
- The pattern of low spreads before crises is broadly present across the alternative dating systems.
- The timing and magnitude differ somewhat because crisis dates are inherently noisy.

Read Table XVII:

- Table XVII repeats the crisis-prediction regressions using Reinhart-Rogoff and Baron-Verner-Xiong dates.
- The froth variables remain informative, especially for panic-style crisis chronologies.
- The results confirm that high credit growth and low spreads are useful warning indicators.

Economic message: The paper's precrisis froth evidence survives major alternative crisis-date definitions.

Table XVI
Spreads Too Low, Alternative Crisis Dates

Spreads too low using alternative crisis dates from Reinhart and Rogoff (2009) and Baron, Verner, and Xiong (2021). We include country and time fixed effects, along with GDP growth and lagged GDP growth as controls, as discussed in Table VII. Driscoll-Kraay standard errors with eight lags are in parentheses.

Variables	(1) JST	(2) RR	(3) BVX
1_{t+5}	-0.13 (0.23)	-0.02 (0.10)	0.01 (0.18)
1_{t+4}	-0.11 (0.21)	-0.01 (0.12)	0.11 (0.17)
1_{t+3}	0.33 (0.28)	0.23 (0.31)	0.51 (0.30)
1_{t+2}	0.25 (0.19)	0.10 (0.16)	0.49 (0.22)
1_{t+1}	0.38 (0.24)	0.17 (0.18)	0.53 (0.15)
1_t	0.50 (0.40)	0.16 (0.14)	0.31 (0.13)
1_{t-1}	0.08 (0.42)	0.10 (0.11)	-0.09 (0.26)
1_{t-2}	-0.50 (0.58)	-0.05 (0.13)	-0.61 (0.46)
1_{t-3}	-0.28 (0.27)	-0.15 (0.10)	-0.35 (0.26)
$Crisis_{t-6,t-4}$	-0.44 (0.15)		
$RR_{t-6,t-4}$		-0.15 (0.08)	
$BVX_{t-6,t-4}$			-0.12 (0.09)
Observations	802	814	814
R^2	0.48	0.45	0.47
Number of groups	16	16	16
Year FE	Y	Y	Y

Table XVI: Spreads too low with alternative dates.

7 Short summary

- The paper studies how credit spreads, credit growth, and GDP behave around financial crises.
- It constructs a long-run international credit-spread panel across 17 countries.
- Financial crises are marked by sharp increases in spreads and persistent declines in GDP.
- The severity of a crisis is best predicted by the interaction between spread increases and high precrisis credit growth.
- Credit growth is interpreted as a proxy for financial fragility.
- Spread increases are interpreted as crisis shocks or credit losses.
- The combination of high fragility and a large shock fits the FZ model of financial crises.
- Before crises, spreads are often unusually low even as credit growth is high.
- This low-spread, high-credit state is labeled credit-market froth.
- Froth predicts financial crises better than high credit growth or low spreads alone.
- Froth predicts financial crises more clearly than ordinary recessions.
- The evidence suggests that credit-supply expansions and underpriced risk are central to crisis run-ups.
- Robustness checks use postwar data, out-of-sample indicators, and alternative crisis chronologies.

Table XVII
Predicting Crises, Alternative Crisis Dates

Crisis Predictions using alternative crisis dates from Reinhart and Rogoff (2009) (Panel A, "RR") and Baron, Verner, and Xiong (2021) (Panel B, "BVX"). Controls are the same as in Table VIII. Driscoll-Kraay standard errors with eight lags are in parentheses.

Panel A: RR				
Variables	(1)	(2)	(3)	(4)
$(HighFroth)_{t,t}$	0.08 (0.05)			-0.13 (0.10)
$(HighCredit)_{t,t}$		0.20 (0.05)		0.02 (0.08)
$(HighFroth) \times (HighCredit)_{t,t}$			0.30 (0.07)	0.38 (0.15)
Observations	678	853	608	608
R^2	0.08	0.09	0.11	0.12
Number of groups	16	15	14	14
Panel B: BVX				
Variables	(1)	(2)	(3)	(4)
$(HighFroth)_{t,t}$	0.09 (0.05)			0.05 (0.08)
$(HighCredit)_{t,t}$		0.20 (0.06)		0.14 (0.09)
$(HighFroth) \times (HighCredit)_{t,t}$			0.15 (0.07)	-0.00 (0.18)
Observations	678	853	608	608
R^2	0.07	0.13	0.09	0.10
Number of groups	16	15	14	14

Table XVII: Predicting crises with alternative dates.

8 Conclusion

8.1 Core conclusion

Krishnamurthy and Muir show that financial crises have a distinctive credit cycle. Credit expands and spreads fall before the crisis, spreads jump sharply when the crisis begins, and output falls persistently afterward. The most severe crises occur when a large spread shock hits after a credit boom.

8.2 Economic intuition

The evidence is naturally summarized by a fragility-times-shock view. A spread spike can occur without a deep recession if the financial system is not fragile. Similarly, high credit growth alone need not produce a crisis. The most damaging episodes combine both: high precrisis fragility and a large credit-market shock.

8.3 Why low spreads matter

Low spreads before crises are important because they suggest that markets are not fully pricing the risks created by credit growth. The boom looks safe precisely when risk is building. This is why the paper emphasizes credit-market froth rather than credit growth alone.

8.4 Implications for macrofinance models

The findings imply that crisis models must explain two facts simultaneously. First, shocks are amplified when the financial sector is fragile. Second, fragility often builds during periods of low spreads, when investors appear optimistic or risk premia are compressed. Models that only generate rising spreads as leverage rises miss the precrisis evidence.

8.5 Policy implications

Credit booms are especially concerning when they coincide with compressed spreads. Monitoring credit growth alone may miss important information, and monitoring spreads alone may also miss danger because spreads can be low before crises. The joint signal is the key warning indicator.

8.6 Limitations

The paper is not a randomized causal design. It relies on historical data, crisis chronologies, and constructed spread measures. Crisis timing is imperfect, and credit growth is an imperfect proxy for financial fragility. Nonetheless, the results are consistent across many specifications and robustness checks.

8.7 Takeaway

The enduring takeaway is that crises are born in tranquil credit markets. Low spreads and rapid credit growth signal that credit supply is expanding and risk is being underpriced. When a shock arrives, the resulting spread spike is most damaging in economies that entered the crisis with high credit fragility.